

Q1 JEE Main 2020 - 7 Jan (Morning)

Let $y = f(x)$ is the solution of the differential equation $e^y \left(\frac{dy}{dx} - 1 \right) = e^x$ such that $y(0) = 0$, then $y(1)$ is equal to:

- (A) $\ln 2$
- (B) $2 + \ln 2$
- (C) $1 + \ln 2$
- (D) $3 + \ln 2$

Q2 JEE Main 2020 - 7 Jan (Evening)

Let $y = y(x)$ be the solution curve of the differential equation, $(y^2 - x) \frac{dy}{dx} = 1$, satisfying $y(0) = 1$. This curve intersects the x -axis at a point whose abscissa is:

- (A) $2 - e$
- (B) $2 + e$
- (C) 2
- (D) e

Q3 JEE Main 2020 - 8 Jan (Morning)

Let $y = y(x)$ be a solution of the differential equation, $\sqrt{1 - x^2} \frac{dy}{dx} + \sqrt{1 - y^2} = 0$, $|x| < 1$. If $y\left(\frac{1}{2}\right) = \frac{\sqrt{3}}{2}$, then $y\left(\frac{-1}{\sqrt{2}}\right)$ is equal to :

- (A) $-\frac{1}{\sqrt{2}}$
- (B) $\frac{\sqrt{3}}{2}$
- (C) $\frac{2}{1}$
- (D) $\frac{1}{\sqrt{2}}$

Q4 JEE Main 2020 - 8 Jan (Evening)

The differential equation of the family of curves, $x^2 = 4b(y + b)$, $b \in R$, is

- (A) $x \left(\frac{dy}{dx} \right)^2 = 2y \frac{dy}{dx} + x^2$
- (B) $x \left(\frac{dy}{dx} \right)^2 = 2y \frac{dy}{dx} + x$
- (C) $x \left(\frac{dy}{dx} \right)^2 = y \frac{dy}{dx} + x^2$

(D) $x \left(\frac{dy}{dx} \right)^2 = y \frac{dy}{dx} + 2x^2$

Q5 JEE Main 2020 - 9 Jan (Morning)

If for $x \geq 0$, $y = y(x)$ is the solution of the differential equation,

$(1+x) \frac{dy}{dx} = ((1+x)^2 + (y-3))$, If $y(2) = 0$ then $y(3) = ?$

Q6 JEE Main 2020 - 9 Jan (Evening)

If $\frac{dy}{dx} = \frac{xy}{x^2+y^2}$; $y(1) = 1$; then a value of x satisfying $y(x) = e$ is:

- (A) $\sqrt{3}e$
- (B) $\frac{e}{\sqrt{2}}$
- (C) $\sqrt{2}e$
- (D) $\frac{1}{2}\sqrt{3}e$

Q7 JEE Main 2020 - 2 Sep (Morning)

Let $y = y(x)$ be the solution of the differential equation, $\frac{2+\sin x}{y+1} \cdot \frac{dy}{dx} = -\cos x$, $y > 0$, $y(0) = 1$.
If $y(\pi) = a$ and $\frac{dy}{dx}$ at $x = \pi$ is b , then the ordered pair (a, b) is equal to :

- (A) (2,1)
- (B) (1,-1)
- (C) $(2, \frac{3}{2})$
- (D) (1,1)

Q8 JEE Main 2020 - 2 Sep (Evening)

If a curve $y = f(x)$, passing through the point $(1, 2)$, is the solution of the differential equation,
 $2x^2 dy = (2xy + y^2) dx$, then $f\left(\frac{1}{2}\right)$ is equal to :

- (A) $\frac{1}{1+\log_e 2}$
- (B) $\frac{-1}{1+\log_e 2}$
- (C) $1 + \log_e 2$
- (D) $\frac{1}{1-\log_e 2}$

Q9 JEE Main 2020 - 3 Sep (Morning)

The solution curve of the differential equation, $(1 + e^{-x})(1 + y^2) \frac{dy}{dx} = y^2$, which passes through the point $(0, 1)$, is :

- (A) $y^2 + 1 = y \left(\log_e \left(\frac{1+e^{-x}}{2} \right) + 2 \right)$
- (B) $y^2 + 1 = y \left(\log_e \left(\frac{1+e^x}{2} \right) + 2 \right)$
- (C) $y^2 = 1 + y \log_e \left(\frac{1+e^{-x}}{2} \right)$
- (D) $y^2 = 1 + y \log_e \left(\frac{1+e^x}{2} \right)$

Q10 JEE Main 2020 - 3 Sep (Evening)

If $x^3 dy + xy dx = x^2 dy + 2y dx$; $y(2) = e$ and $x > 1$, then $y(4)$ is equal to :

- (A) $\frac{\sqrt{e}}{2}$
- (B) $\frac{1}{2} + \sqrt{e}$
- (C) $\frac{3}{2} + \sqrt{e}$
- (D) $\frac{3}{2} \sqrt{e}$

Q11 JEE Main 2020 - 4 Sep (Morning)

Let $y = y(x)$ be the solution of the differential equation, $xy' - y = x^2(x \cos x + \sin x)$, $x > 0$. If $y(\pi) = \pi$, then $y''\left(\frac{\pi}{2}\right) + y\left(\frac{\pi}{2}\right)$ is equal to :

- (A) $1 + \frac{\pi}{2}$
- (B) $1 + \frac{\pi}{2} + \frac{x^2}{4}$
- (C) $2 + \frac{\pi}{2}$
- (D) $2 + \frac{\pi}{2} + \frac{\pi^2}{4}$

Q12 JEE Main 2020 - 4 Sep (Evening)

The solution of the differential equation $\frac{dy}{dx} - \frac{y+3x}{\log_e(y+3x)} + 3 = 0$ is :

(where C is a constant of integration.)

- (A) $x - \frac{1}{2}(\log_e(y+3x))^2 = C$
- (B) $y + 3x - \frac{1}{2}(\log_e x)^2 = C$
- (C) $x - 2 \log_e(y+3x) = c$
- (D) $x - \log_e(y+3x) = c$

Q13 JEE Main 2020 - 5 Sep (Morning)

If $y = y(x)$ is the solution of the differential equation $\frac{5+e^x}{2+y} \cdot \frac{dy}{dx} + e^x = 0$ satisfying $y(0) = 1$ then a value of $y(\log_e 13)$ is :

- (A) -1
- (B) 2
- (C) 0
- (D) 1

Q14 JEE Main 2020 - 5 Sep (Evening)

Let $y = y(x)$ be the solution of the differential equation

$\cos x \frac{dy}{dx} + 2y \sin x = \sin 2x, x \in (0, \frac{\pi}{2})$. If $y(\pi/3) = 0$, then $y(\pi/4)$ is equal to :

- (A) $\frac{1}{\sqrt{2}} - 1$
- (B) $\sqrt{2} - 2$
- (C) $2 - \sqrt{2}$
- (D) $2 + \sqrt{2}$

Q15 JEE Main 2020 - 6 Sep (Morning)

The general solution of the differential equation

$\sqrt{1+x^2+y^2} + x^2y^2 + xy \frac{dy}{dx} = 0$ is

(where C is a constant of integration)

- (A) $\sqrt{1+y^2} + \sqrt{1+x^2} = \frac{1}{2} \log_e \left(\frac{\sqrt{1+x^2+1}}{\sqrt{1+x^2-1}} \right) + C$
- (B) $\sqrt{1+y^2} + \sqrt{1+x^2} = \frac{1}{2} \log_e \left(\frac{\sqrt{1+x^2+1}}{\sqrt{1+x^2+1}} \right) + C$
- (C) $\sqrt{1+y^2} - \sqrt{1+x^2} = \frac{1}{2} \log_e \left(\frac{\sqrt{1+x^2-1}}{\sqrt{1+x^2+1}} \right) + C$
- (D) $\sqrt{1+y^2} - \sqrt{1+x^2} = \frac{1}{2} \log_e \left(\frac{\sqrt{1+x^2+1}}{\sqrt{1+x^2-1}} \right) + c$

Q16 JEE Main 2020 - 6 Sep (Evening)

If $y = \left(\frac{2}{\pi}x - 1\right) \operatorname{cosec} x$ is the solution of the differential equation, $\frac{dy}{dx} + p(x)y = \frac{2}{\pi} \operatorname{cosec} x$ $0 < x < \frac{\pi}{2}$, then the function $p(x)$ is equal to :

- (A) $\tan x$
- (B) $\cot x$

(C) $\operatorname{cosec} x$

(D) $\sec x$

Answer Key

Q1 (C)	Q2 (A)	Q3 (C)	Q4 (B)
Q5 (3)	Q6 (A)	Q7 (D)	Q8 (A)
Q9 (D)	Q10 (D)	Q11 (C)	Q12 (A)
Q13 (A)	Q14 (B)	Q15 (A)	Q16 (B)

Q1

$$e^y = t$$

$$e^y \frac{dy}{dx} = \frac{dt}{dx}$$

$$\frac{dt}{dx} - t = e^x$$

$$I.F. = e^{\int -1 dx} = e^{-x}$$

$$t(e^{-x}) = \int e^x \cdot e^{-x} dx$$

$$e^{y-x} = x + c$$

Put $x = 0$, $y = 0$ then $c = 1$

$$e^{y-x} = x + 1$$

$$y = x + \ln(x + 1)$$

$$\text{at } x = 1 \quad y = 1 + \ln(2)$$

Q2

$$\frac{dx}{dy} + x = y^2$$

$$I.F. = e^{\int 1 dy} = e^y$$

$$x \cdot e^y = \int y^2 \cdot e^y \cdot dy$$

$$= y^2 \cdot e^y - \int 2y \cdot e^y \cdot dy$$

$$\Rightarrow y^2 e^y - 2(y \cdot e^y - e^y) + c$$

$$x \cdot e^y = y^2 e^y - 2ye^y + 2e^y + c$$

$$x = y^2 - 2y + 2 + c \cdot e^{-y}$$

$$x = 0, \quad y = 1$$

$$0 = 1 - 2 + 2 + \frac{c}{e}$$

$$c = -e$$

$$y = 0, \quad x = 0 - 0 + 2 + (-e)(e^{-0})$$

$$x = 2 - e$$

Q3

$$\frac{dy}{\sqrt{1-y^2}} + \frac{dx}{\sqrt{1-x^2}} = 0 \Rightarrow \sin^{-1}y + \sin^{-1}x = c$$

$$\text{At } x = \frac{1}{2}, y = \frac{\sqrt{3}}{2} \Rightarrow c = \frac{\pi}{2} \Rightarrow \sin^{-1}y = \cos^{-1}x$$

$$\text{Hence } y\left(\frac{-1}{\sqrt{2}}\right) = \sin\left(\cos^{-1}\frac{-1}{\sqrt{2}}\right) = \frac{1}{\sqrt{2}}$$

Q4

$$2x = 4by' \Rightarrow b = \frac{x}{2y'}$$

$$\text{So, differential equations is } x^2 = \frac{2x}{y'} \cdot y + \left(\frac{x}{y'}\right)^2$$

$$\Rightarrow x\left(\frac{dy}{dx}\right)^2 = 2y\frac{dy}{dx} + x$$

Q5

$$\frac{dy}{dx} = (1+x) + \left(\frac{y-3}{1+x}\right)$$

$$\frac{dy}{dx} - \frac{1}{(1+x)}y = (1+x) - \frac{3}{(1+x)}$$

$$\text{I.F.} = e^{-\int \frac{1}{(1+x)}dx} = \frac{1}{(1+x)}$$

$$\therefore \frac{d}{dx}\left(\frac{y}{1+x}\right) = 1 - \frac{3}{(1+x)^2}$$

$$\frac{y}{1+x} = x + 3(1+x)^{-1} + c$$

$$y = (1+x)\left[x + \frac{3}{(1+x)} + c\right]$$

$$\therefore \text{ at } x = 2, y = 0$$

$$\therefore 0 = 3(2+1+c) \Rightarrow c = -3$$

$$\therefore \text{ at } x = 3, y = 3$$

Q6

Put $y = vx$

$$\frac{dy}{dx} = v + x \frac{dv}{dx}$$

$$v + x \frac{dv}{dx} = \frac{vx^2}{x^2 + v^2x^2}$$

$$\Rightarrow \frac{1+v^2}{v^3} dv = -\frac{1}{x} dx$$

$$\Rightarrow \int \left(\frac{1}{v^3} + \frac{1}{v} \right) dv = \int -\frac{1}{x} dx$$

$$\Rightarrow \frac{-1}{2} \frac{1}{v^2} + \ln v = -\ln x + c$$

$$\Rightarrow -\frac{x^2}{2y^2} = -\ln y + c$$

When $x = 1$, $y = 1$ then

$$-\frac{1}{2} = c$$

$$\Rightarrow x^2 = y^2 (1 + 2\ln y)$$

$$\Rightarrow x^2 = e^2 (3)$$

$$\Rightarrow x = \pm \sqrt{3}e$$

So $x = \sqrt{3}e$

Q7

$$\frac{2+\sin x}{y+1} \frac{dy}{dx} = -\cos x, y > 0$$

$$\int \frac{dy}{y+1} = \int \frac{(-\cos x)dx}{2+\sin x}$$

$$\ln |y+1| = -\ln |2+\sin x| + \ln c$$

$$\ln |(y+1)(2+\sin x)| = \ln c$$

$$\therefore y(0) = 1 \Rightarrow \ln 4 = \ln c \Rightarrow c = 4$$

$$\therefore (y+1)(2+\sin x) = 4$$

$$\therefore y = \frac{2-\sin x}{2+\sin x}$$

$$\therefore a = y(\pi) = 1$$

and $\frac{dy}{dx} = \frac{(2+\sin x)(-\cos x) - (2-\sin x) \cdot \cos x}{(2+\sin x)^2}$

$$\left. \frac{dy}{dx} \right|_{x=\pi} = 1 = b$$

Q8

$$\frac{dy}{dx} = \frac{2xy+y^2}{2x^2}$$

Put $y = vx$

$$v + x \frac{dv}{dx} = v + \frac{v^2}{2}$$

$$\Rightarrow 2 \frac{dv}{v^2} = \frac{dx}{x}$$

$$\Rightarrow \frac{-2}{v} = \ln x + c$$

$$\Rightarrow \frac{-2x}{y} = \ln x + c$$

$$\Rightarrow (1, 2)$$

$$\Rightarrow \frac{c-2x^{-1}}{y} = \ln x - 1$$

$$\text{Hence, for } x = \frac{1}{2} \Rightarrow y = \frac{1}{1+\ln 2}$$

Q9

$$\int \left(\frac{y^2+1}{y^2} \right) dy = \int \frac{e^x dx}{e^x+1}$$

$$\Rightarrow y - \frac{1}{y} = \ln|e^x+1| + c$$

$$\downarrow (0, 1)$$

$$c = -\ln 2$$

$$\Rightarrow y^2 - 1 = y \ln \left(\frac{e^x+1}{2} \right)$$

$$\Rightarrow y^2 = 1 + y \ln \left(\frac{e^x+1}{2} \right)$$

Q10

$$(x^3 - x^2) dy = y(2 - x) dx$$

$$\Rightarrow \int \frac{dy}{y} = \int \frac{2-x}{x^2(x-1)} dx \quad (1)$$

Making partial fractions for RHS

$$\frac{2-x}{x^2(x-1)} = \frac{A}{x} + \frac{B}{x^2} + \frac{C}{x-1}$$

$$\therefore A = -1, B = -2, C = 1$$

$$\Rightarrow \int \frac{dy}{y} = \int \left(\frac{-1}{x} - \frac{2}{x^2} + \frac{1}{x-1} \right) dx$$

$$\Rightarrow \ln y = -\ln x + \frac{2}{x} + \ln(x-1) + c$$

$$\text{as } y(2) = e \Rightarrow c = \ln 2$$

$$\Rightarrow y(4) = \frac{3}{2} \sqrt{e}$$

Q11

$$\frac{dy}{dx} - \frac{y}{x} = x(x \cos x + \sin x)$$

$$1. F = e^{-\int \frac{1}{x} dx} = \frac{1}{x}$$

$$\Rightarrow \int d\left(\frac{y}{x}\right) = \int (x \cos x + \sin x) dx$$

$$\Rightarrow \frac{y}{x} = x \sin x + c$$

$$\text{Also } y(\pi) = \pi \Rightarrow c = 1$$

$$y = x^2 \sin x + x \Rightarrow y\left(\frac{\pi}{2}\right) = \frac{\pi^2}{4} + \frac{\pi}{2}$$

$$y' = 2x \sin x + x^2 \cos x + 1$$

$$y'' = 2 \sin x - x^2 \sin x \Rightarrow y''\left(\frac{\pi}{2}\right) = 2 - \frac{\pi^2}{4}$$

$$\therefore y\left(\frac{\pi}{2}\right) + y''\left(\frac{\pi}{2}\right) = 2 + \frac{\pi}{2}$$

Q12

$$\text{Let } y + 3x = t$$

$$\Rightarrow \frac{dy}{dx} + 3 = \frac{dt}{dx}$$

$$\text{Then } \frac{dt}{dx} = \frac{t}{\ln t}$$

$$\Rightarrow \frac{\ln t}{t} dt = dx$$

$$\Rightarrow \frac{(\ln t)^2}{2} = x - C$$

$$\Rightarrow x - \frac{1}{2}(\ln(y + 3x))^2 = C$$

Q13

$$\frac{5+e^x}{2+y} \cdot \frac{dy}{dx} + e^x = 0$$

$$\int \frac{dy}{2+y} = - \int \frac{e^x}{5+e^x} dx$$

$$\ln|2+y| + \ln|5+e^x| = \ln C$$

$$\therefore y(0) = 1$$

$$\Rightarrow \ln C = \ln 18$$

$$\therefore |(2+y) \cdot (5+e^x)| = 18$$

$$\text{When } x = \ln 13 \text{ then } |(2+y) \cdot 18| = 18$$

$$2+y = \pm 1$$

$$\therefore y = -1, -3$$

$$\therefore y(\ln 13) = -1$$

Q14

$$\frac{dy}{dx} + 2y \tan x = 2 \sin x$$

$$\text{I.F.} = e^{\int 2 \tan x dx} = \sec^2 x$$

$$\Rightarrow y \cdot \sec^2 x = \int 2 \sin x \cdot \sec^2 x dx + c$$

$$\Rightarrow y \sec^2 x = 2 \sec x + c$$

$$\text{When } x = \frac{\pi}{3}, y = 0; \text{ then } c = -4$$

$$\Rightarrow y \sec^2 x = 2 \sec x - 4 \Rightarrow y = \frac{2 \sec x - 4}{\sec^2 x}$$

$$\Rightarrow y\left(\frac{\pi}{4}\right) = \sqrt{2} - 2$$

Q15

$$\sqrt{1+x^2+y^2+x^2y^2} + xy \frac{dy}{dx} = 0$$

$$\sqrt{1+x^2} \cdot \sqrt{1+y^2} = -xy \frac{dy}{dx}$$

$$\int \frac{\sqrt{1+x^2}}{x} dx = - \int \frac{y}{\sqrt{1+y^2}} dy$$

$$\text{Let } I = \int \frac{\sqrt{1+x^2}}{x} dx \text{ Let } x = \tan \theta \Rightarrow dx = \sec^2 \theta d\theta$$

$$= \int \frac{\sec^3 \theta d\theta}{\tan \theta} = \int \frac{\sin^2 \theta + \cos^2 \theta}{\sin \theta \cdot \cos^2 \theta} d\theta$$

$$= \int (\tan \theta \cdot \sec \theta + \operatorname{cosec} \theta) d\theta$$

$$= \sec \theta + \ln |\operatorname{cosec} \theta - \cot \theta| + C$$

$$= \sqrt{1+x^2} + \ln \left| \frac{\sqrt{1+x^2}-1}{x} \right| + C$$

$$\sqrt{1+x^2} + \ln \left| \frac{\sqrt{1+x^2}-1}{x} \right| = -\sqrt{1+y^2} + C$$

$$\sqrt{1+y^2} + \sqrt{1+x^2} = \ln \left| \frac{x}{\sqrt{1+x^2}-1} \right| +$$

$$C = \frac{1}{2} \ln \left(\frac{\sqrt{1+x^2}+1}{\sqrt{1+x^2}-1} \right) + C$$

Q16

$$\because y = \left(\frac{2}{\pi}x - 1\right) \operatorname{cosec} x$$

$$\frac{dy}{dx} = \frac{2}{\pi} \operatorname{cosec} x - \left(\frac{2}{\pi}x - 1\right) \operatorname{cosec} x \cdot \cot x$$

$$= \operatorname{cosec} x \left[\frac{2}{\pi} - \left(\frac{2}{\pi}x - 1\right) \cot x \right]$$

$$\Rightarrow \frac{dy}{dx} - \frac{2}{\pi} \operatorname{cosec} x = -yp(x)$$

$$\Rightarrow yp(x) = \left(\frac{2}{\pi}x - 1\right) \cot x \cdot \operatorname{cosec} x$$

$$\Rightarrow p(x) = \cot x$$

